Sai Kiran Munugoti

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SUMMARY

Data Scientist with 7+ years of experience building predictive models, quantitative risk frameworks, and analytical pipelines across financial services, insurance, and operations. Core work spans multi-source data integration, machine learning model development and validation, and translating outputs into prioritization tools and dashboards for operational and executive decision-making. Experienced working in cross-functional environments where models must be explainable, traceable, and tied to measurable business outcomes.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), R, SQL

Machine Learning: Logistic Regression, Random Forest, XGBoost, Gradient Boosting, Survival Analysis, Bayesian Methods, K-Means Clustering, Neural Networks

Risk & Analytics: Quantitative risk scoring, event likelihood/consequence modeling, anomaly detection, time-to-event modeling, scenario analysis

Forecasting: ARIMA, Prophet, time-series modeling, feature engineering, hyperparameter tuning

Data Engineering: ETL pipeline development, data quality validation, Azure Data Factory, AWS Redshift, Snowflake, SSIS, Talend, PySpark, PostgreSQL

Visualization: Power BI, Tableau, Matplotlib, Seaborn, RShiny, QuickSight

Geospatial: ArcGIS, GeoPandas, spatiotemporal analysis

Cloud & MLOps: AWS SageMaker, Azure, Google Cloud, model validation, drift monitoring, Great Expectations

Other: JIRA, Agile methodology, Blue Prism RPA, Salesforce, SAP, API Integration, Data Parsing

PROJECTS

HST Ingredient Tracker — *Personal Project, 2026*

A food and cosmetics hazard-substance scanner: search any product to see additive analysis, sugar profile, and hazard alerts, backed by a shared community database.

PostgreSQL · API Integration · Data Parsing · Ingredient/Additive Taxonomy

PROFESSIONAL EXPERIENCE

Data Scientist – RB Management LLC *Dec 2025 – Present Remote, USA*

- Designed multi-dimensional risk scoring models integrating operational, financial, and behavioral data from hotel PMS, POS, CRM, and booking platforms using Azure Data Factory and SQL.
- Built anomaly detection and forecasting models to identify emerging risk patterns across a multi-property hotel portfolio, enabling proactive operational interventions.
- Developed Power BI dashboards that ranked risk priorities and supported resource allocation decisions for Operations, Revenue Management, and Finance leadership.
- Established data quality governance standards and validation controls across all data pipelines, improving downstream analytics reliability.
- Partnered with cross-functional leadership to convert ambiguous operational challenges into structured analytical problem statements.

Data Scientist – Cognitum Solutions LLC *Apr 2025 – Dec 2025 Remote, USA*

- Built time-series forecasting models using ARIMA and Prophet to support financial risk planning; improved forecast accuracy through iterative feature engineering and model calibration.
- Conducted root-cause analysis on security incidents and produced visual risk summaries for executive review, including trend identification and anomaly flagging.
- Developed ensemble ML strategies (stacking, blending) to improve model robustness on heterogeneous, incomplete datasets from multiple operational systems.
- Supported design of risk mitigation control procedures and operational playbooks aligned with compliance requirements.

- Delivered stakeholder training on dashboard interpretation, risk metric definitions, and data-driven workflow adoption.

Data Scientist – Protek Talent Inc *Feb 2024 – Apr 2025 Remote, USA*

- Built customer churn prediction models using Logistic Regression and XGBoost, achieving 87% accuracy on noisy, multi-source operational data.
- Automated ETL workflows for financial data ingestion, transformation, and validation, reducing pipeline processing time by 40% and improving risk analysis throughput.
- Applied K-Means clustering and RFM analysis to segment customer risk profiles for targeted retention and mitigation strategies.
- Resolved model overfitting through Grid Search and Random Search hyperparameter tuning, improving generalization performance by 20% on held-out test sets.
- Debugged inconsistent model behavior by validating data schema changes using Pandas Profiling and automated pipeline checks with Great Expectations.

Data Scientist – Genpact India Pvt. Ltd. (Cigna Healthcare) *Jun 2021 – Aug 2022 Bengaluru, India*

- Developed predictive models to identify high-risk insurance claims and potential fraud patterns, contributing to a 68% improvement in fraud detection rates and 30% reduction in associated operational costs.
- Integrated multi-source healthcare data (claims, provider networks, member services) via SQL and Azure Data Factory, improving data quality and downstream analytics reliability for compliance reporting.
- Applied statistical anomaly detection to flag inconsistent claims data; collaborated with actuaries and compliance teams on root-cause analysis and regulatory remediation.
- Automated 70% of document-intensive data collection workflows using Blue Prism RPA integrated with NLP document processing.
- Optimized insurance database schema design, increasing claims processing efficiency by 20%.

Data Analyst – SequelSoft India Limited *Apr 2018 – Jun 2021 Hyderabad, India*

- Built ETL pipelines using Talend and AWS Redshift, improving data transformation accuracy and reporting reliability by 25% across financial and operational data domains.
- Developed Power BI and Tableau dashboards to support business performance tracking across inventory, sales, and returns reporting.
- Optimized complex SQL queries by analyzing execution plans and restructuring joins, subqueries, and indexes, achieving up to 50% reduction in query execution time.
- Supported migration of on-premises databases to AWS and Google Cloud, ensuring security compliance and data integrity throughout transition.
- Developed reconciliation scripts to identify staging-to-production data mismatches using checksums and join validation.

Junior Data Analyst – Henceforth Technologies *Jun 2015 – Mar 2018 Hyderabad, India*

- Improved database query performance and data processing efficiency by 30% through storage optimization and query restructuring.
- Consolidated data from multiple source systems into BI tools (Tableau, Power BI), ensuring accuracy and consistency for business reporting.
- Developed SSIS packages for data import and export across Text, Excel, and CSV formats to SQL Server tables; implemented error handling and event-driven notifications.
- Conducted data validation and cleansing to maintain integrity and compliance with security and governance policies.

EDUCATION

Ph.D., Artificial Intelligence – University of Cumberlands (in progress)

M.S., Computer Science and Information Technology Management – Concordia University, Saint Paul, MN

B.Tech, Mechanical Engineering (Minor: Computer Science) – Tirumala Engineering College, JNTU-K, India

CERTIFICATIONS

- Machine Learning with R – IBM Cognitive Class (IBM, 2019)
- Certified RPA Developer, Blue Prism Accredited Developer AD01 (Blue Prism, 2018)